Manuscript Outline & Key Points

Nucleotide-resolution prediction of ribosome occupancy from transcriptome sequences across cellular contexts

INTRODUCTION

1. Translation varies across transcripts and cellular contexts, so transcript abundance alone cannot predict protein output. Nucleotide-resolution occupancy models could support both sequence design and translated-ORF discovery.
2. Ribo-seq provides nucleotide-resolution occupancy but is difficult to deploy broadly. Existing models usually separate TE estimation, local dynamics and ORF identification, and often require CDS annotations, fixed windows or Ribo-seq input.
3. A unified model must capture multiscale sequence features and cellular context while learning from heterogeneous Ribo-seq data. We address this with harmonized targets, variable-length modeling, context conditioning and chromosome-held-out evaluation.
4. TRACE predicts full-length occupancy profiles and is annotation-free at inference. Trained on 1.8 million transcript–context profiles from 73 cellular contexts across three species, it unifies translation dynamics, TE, ORF identification, cultured-cell design and neoantigen prioritization (Supplementary Table 3).

RESULT 1: A nucleotide-resolution dataset of ribosome occupancy across cellular contexts

1. Raw Ribo-seq signal is sparse, protocol dependent and confounded by transcript abundance.
2. We curated 1,355 Ribo-seq runs and 475 matched RNA-seq runs from 73 cellular contexts across human, rhesus macaque and mouse.
3. A unified workflow performed quality control, within-context merging, representative-transcript selection, nuclease-aware P-site inference, profile filtering and RNA-abundance normalization.
4. Within-context merging substantially increased per-transcript coverage, with CDS coverage of housekeeping genes approaching 1.0.
5. Compositional regression reduced RNA-abundance confounding to yield relative ribosome occupancy profiles.
6. Cross-species and cross-context patterns are consistent with a substantial cis-sequence contribution without establishing causal dominance. This supports an RNA-sequence backbone with cellular context conditioning.
7. Processed profiles were enriched across annotated CDSs and showed strong three-nucleotide periodicity.
8. The final resource contains 1.8 million transcript–context profiles, with one profile for each observed transcript and cellular context (Fig. 1).

9. The resource includes protein-coding transcripts and noncoding RNAs with measurable occupancy but does not represent every translatome state.

RESULT 2: TRACE predicts nucleotide-resolution ribosome occupancy profiles

1. TRACE uses full-length RNA sequence as a shared backbone; bidirectional attention, Rotary Position Embedding, Flash Attention and length-aware batching support variable-length transcripts.
2. Adaptive Layer Normalization conditions the shared sequence backbone on cellular context.
3. Training combines position-level occupancy, CDS frame-aware TE and batchwise transcript-ranking objectives. CDS and frame labels supervise training, whereas inference requires no annotations.
4. On chromosome-held-out transcripts, TRACE produced periodic, CDS-enriched occupancy profiles, indicating generalization of supervised translation patterns (Fig. 2b,c).
5. Predicted periodicity differed between housekeeping transcripts and noncoding RNAs (Fig. 2d).
6. For ENST00000332859, predicted and observed profiles reached Spearman $\rho = 0.80$. A localized noncoding-RNA prediction overlapped Ribo-seq evidence without establishing stable protein production.
7. The same profile supports TE estimation and forward design within known CDSs and annotation-free ORF discovery across unannotated transcripts.

RESULT 3: A unified translation representation supports three benchmark tasks

1. TRACE was benchmarked on translation dynamics, TE and ORF identification. Among the models in Supplementary Table 3, only TRACE combines nucleotide-resolution occupancy, cellular context conditioning, unified multi-species modeling and annotation-free inference.
2. For translation dynamics, TRACE approached cross-study experimental baselines and outperformed retrained sequence-only Riboformer and RiboMIMO on the evaluated datasets (Fig. 3b–d).
3. For TE, TRACE ranked among the top-performing models across polysome profiling, protein-to-RNA ratios and SILAC synthesis rates (Fig. 3e–g).
4. For ORF identification, TRACE achieved higher F1 than the evaluated Ribo-seq-based callers and sequence baselines without sample-specific Ribo-seq input.
5. In matched fetal-brain data, TRACE profiles correlated with Ribo-seq, and short-ORF precision was comparable to RiboTIE.

RESULT 4: TRACE predictions show translation-associated patterns across sequence scales

1. Prediction sensitivity and attention were analyzed across transcript architecture, initiation motifs and codon positions. Because CDS and frame labels supervised training, attention is interpreted associatively rather than causally.
2. Predicted TE was associated with 5′ UTR, CDS and 3′ UTR length and with RNA minimum free energy; these correlations do not establish causality.
3. Attention was enriched near start, CDS and stop regions and was associated with Kozak context.
4. Frame-resolved attention reflected training supervision, whereas in silico codon substitutions measured prediction sensitivity rather than elongation mechanisms.
5. In silico upstream-start perturbations produced prediction changes consistent with known initiation-associated patterns.

RESULT 5: Forward design of CDS translation in specific cellular contexts

1. Ten Fluc and mCherry designs showed graded cultured-cell protein output associated with TRACE predictions (Pearson $r > 0.9$).
2. A TRACE-designed Gluc CDS produced higher expression than wild-type, RiboDecode and LinearDesign sequences in the evaluated cultured-cell experiment.
3. Cellular-context-specific designs produced the intended relative expression patterns in HEK293T, HeLa and HepG2 cells.

RESULT 6: Reverse discovery of tumor neoantigen candidates from transcriptomes

1. Non-canonical translation and alternative splicing expand candidate epitopes beyond somatic mutations but require prioritization of translated ORFs.
2. Annotation-free occupancy prediction supports candidate nomination from tumor and normal RNA-seq without matched tumor Ribo-seq or mass spectrometry.
3. The pipeline combined isoform assembly, tumor-specific transcript selection, TRACE ORF prediction and HLA binding prediction to nominate ten neoepitopes.
4. IFN-γ ELISpot and organoid co-culture supported selected candidates functionally but did not directly quantify endogenous peptide presentation.

DISCUSSION

1. TRACE combines a harmonized resource of 1.8 million transcript–context profiles with full-length sequence modeling and cellular context conditioning.
2. Cross-species patterns support sequence-centered modeling, but neither these associations nor supervised attention establish causal mechanisms.
3. One translation representation links forward CDS design with annotation-free reverse discovery of candidate ORFs and neoantigens.

4. Limitations include reduced accuracy for low-occupancy transcripts and underrepresented contexts, the distinction between occupancy and stable protein output, and the absence of direct immunopeptidomic confirmation.
5. The harmonized data representation and joint ORF and TE objectives could provide translation-focused fine-tuning for RNA foundation models.
6. TRACE could guide cell-type-specific expression optimization for protein-replacement mRNAs and Cas proteins, but these applications require in vivo validation.
7. Population-scale tumor-transcriptome analysis could prioritize recurrent neoantigen candidates across patients. Validation across HLA backgrounds may enable more broadly deployable mRNA vaccines and reduce, rather than eliminate, individualized design costs.

FIGURES

Figure	**Content**
Fig. 1	Dataset: nucleotide-resolution ribosome occupancy and cellular context-associated patterns
Fig. 2	TRACE architecture, annotation-free inference on held-out transcripts, CDS enrichment and biotype periodicity
Fig. 3	Unified benchmarks: translation dynamics, TE, ORF identification and matched multi-omics validation
Fig. 4	Associative interpretation: architecture, MFE, attention, Kozak context, codon-position sensitivity and upstream starts
Fig. 5	Forward design: graded reporters, Gluc comparison and expression specific to each cellular context in cultured cells
Fig. 6	Reverse discovery: neoantigen candidate prioritization, IFN-γ ELISpot and organoid co-culture